

LexGraph: Proof-Carrying Precedent Prediction from Facts over Indian Land-Dispute Jurisprudence

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Abstract—Large language models used to answer legal questions routinely fabricate citations and are poor at recognizing their own errors. Most systems that address this treat it as an extraction problem: given a decided judgment, recover its structure. That is a useful annotation task but not the task anyone faces, because a litigant holds facts and evidence, not a finished opinion. We present LexGraph, a pipeline that inverts the direction. Given only the facts of a dispute, it retrieves candidate precedents from a corpus of 5,462 Indian Supreme Court judgments, constrains them to cases that predate the dispute, and requires that every authority named in the generated answer resolve deterministically back to a retrieved case before the answer is allowed to stand; authorities that cannot be certified are refused rather than included. Evaluation labels are free: a judgment’s own citations are held out and used as ground truth, and every sentence carrying a citation signal is stripped from the query so the task cannot leak. On 100 post-2000 query cases, retrieval from facts alone recovers 15.6% of reachable authorities at $k=10$ against 5.3% for a popularity control that always returns the most-cited cases, establishing that the retriever reads the dispute rather than exploiting citation skew. Reducing the query from a whole judgment to 2,000 characters of facts costs only half the recall, indicating that the facts, not the court’s reasoning, carry most of the retrieval signal. Lexical and dense retrieval trade places by query length: dense leads on short fact queries, BM25 on whole judgments. The central result concerns fabrication. Asked for authorities from the facts with no retrieved brief, 46 of 61 proposed citations resolve to no case anywhere in the corpus, a 75% fabrication rate; drafting from a retrieved brief with deterministic certification drives this to zero. We also report the honest limit: only 13.1% of certified authorities were ones the court actually cited, so certification guarantees provenance rather than correctness. A measured property of the corpus bounds all recall figures — only 25.6% of citation edges point at a case the archive contains, because it holds Supreme Court judgments while those judgments cite High Courts, the Privy Council and English decisions.

Index Terms—precedent retrieval, legal citation verification, large language model hallucination, signed graphs, retrieval-augmented generation, neuro-symbolic legal AI

I. Introduction

Large language models are increasingly applied to case law, yet they remain probabilistic text generators in a domain where a wrong citation is not a cosmetic defect. Dahl et al. [1] found that general-purpose models hallucinate legal facts or authorities in the majority of tested queries and are poor at recognizing their own

errors. A fabricated authority, or a real authority cited for a proposition it does not stand for, can misdirect a self-represented litigant, contaminate a filing, or render an administrative decision unreviewable. A system that cannot exhibit which rule it applied and which authority supports it cannot be contested at all.

There is a second failure that receives less attention, and it is about direction. Most work in this area — including the first version of this project — takes a decided judgment as input and recovers typed structure from it: the issue, the rule, the cited authorities, the holding. That is a useful annotation task. It is not, however, the task anyone actually faces. A litigant, a clerk, or a duty lawyer holds facts and evidence and needs to know which rules and which prior decisions apply. A system that works only after a court has written the answer cannot help before the answer exists. Worse, a system evaluated on extraction from a judgment is being scored on a document that already contains its own ground truth, which makes the evaluation easier than the deployed problem.

This paper inverts the direction. The input is the factual narrative of a dispute; the output is the set of authorities that apply, each carrying a proof that it is a real case, is present in the corpus, and predates the dispute. Retrieval supplies coverage, a signed citation graph supplies an assessment of whether a precedent is still good law, and a deterministic certification step supplies the guarantee: an authority that cannot be resolved back to a retrieved case is refused, not included with a hedge.

The contributions are as follows.

- An inverted, leakage-controlled evaluation. A judgment’s own citations are held out as labels, giving thousands of supervised examples at no annotation cost, and every sentence carrying a citation signal is removed from the query so the task cannot be solved by copying. We show that the pre-2000 headnote convention, which announces cited authorities at the top of a judgment, makes those decades unusable as queries — exactly inverting the difficulty profile of extraction-based work.
- A measured ceiling. Only 25.6% of case-citation edges point at a case this Supreme-Court-only corpus contains. We report recall against reachable authorities

with raw totals alongside, so no figure is flattered by a limit that belongs to the archive rather than the method.

- A fabrication result with a control. Given facts and no brief, 75% of proposed authorities resolve to no case in the corpus; drafting from a retrieved brief with deterministic certification drives that to zero. A popularity control establishes that retrieval is reading the dispute, not exploiting the skew of citation graphs.
- A separation of groundedness from correctness. Certification guarantees provenance, not that the right case was chosen: only 13.1% of certified authorities were ones the court actually cited. We argue these are distinct metrics and report both.
- Two diagnosed extraction failures. We show that candidate ordering, not model capability, bounded the earlier extraction pipeline, and that keying extraction rules on the content shape of a span removes all 196 misrouted statute references from the authority channel at no cost in recall.

II. Related Work

Legal knowledge representation has a long history of separating legal concepts from the logic of their application. LKIF-Core [3] provides a modular OWL ontology of basic legal concepts (norms, acts, roles, time), and OASIS LegalRuleML [4] extends this with a machine-readable syntax for obligations, permissions, and defeasible rules. These frameworks show that legal text can be lifted into structured, queryable representations, but they were built by hand and do not scale to large modern corpora.

Sunny and Sivan-Sevilla [7] address this scaling problem directly: they use an LLM to translate statutory eligibility rules into a formal ontology, then use satisfiability-based (SMT) verification to check whether real welfare-benefit explanations are legally adequate. Their central finding — that formal verification catches statutory violations even in explanations that look reasonable to a human reader — is strong evidence that neuro-symbolic verification adds something that formal-logic-free LLM prompting cannot. LexGraph applies the same “neural extraction plus symbolic verification” division of labor to a different artifact (judicial citation reasoning instead of benefits notices) and a different verification target: does a cited authority actually exist and get relied upon, rather than does an explanation satisfy a statute.

Our extraction method follows an older lineage: Riloff [6] introduced AutoSlog, which learns symbolic extraction patterns from examples using generic syntactic templates (active-verb-plus-object, passive-verb-plus-subject, etc.). We replace Riloff’s syntactic patterns with LLM-grounded semantic patterns — the model maps a sentence onto a small ontology of legal roles and actions instead of a parse tree — but keep AutoSlog’s core idea that extraction

should produce inspectable, reusable rules rather than one-off, black-box answers.

On the graph side, citation networks between judgments are naturally a signed ecosystem: a later case can rely on, distinguish, or overrule an earlier one, and treating all edges as equivalent “citation” loses the information that matters most for assessing whether a proposition is still good law. Signed Graph Convolutional Networks [2] show that explicitly modeling positive and negative edges improves representation quality over unsigned-graph baselines; we adapt this idea at a much lighter weight (a heuristic- or classifier-derived sign label plus a structural score) rather than a full SGCN, given the scope of a one-semester course project.

Finally, retrieval-augmented generation [5] is the natural grounding-in-source-text baseline for this task, and we use it as one, rather than only as an ablation of our own pipeline: Section VI shows it substantially reduces, but does not eliminate, the fabrication problem that closed-book prompting exhibits.

III. Dataset

The provided corpus is an archive of 26,688 Indian Supreme Court judgments (1950–2025), year-organized PDF files, plus two locally available derived assets: a citation metadata set covering 7,493 citing cases with 144,353 total outbound citation edges, and a filtered index of 6,954 judgments identified as land or property disputes.

The raw citation metadata mixes case-to-case citations with statute, article, and section references under a single undifferentiated list. A rule-based classifier splits these apart: of 144,353 edges, 32.5% are genuine case citations, 54.4% are statute or article references, and 13.1% are unresolved fragments (mostly bare sub-clause numbers trailing a statute citation, e.g. “(7)”), left for manual review rather than forced into a bucket. Every downstream use of a citation edge in this paper goes through this classifier first; treating the raw “precedents” field as case citations would silently double the apparent citation count with statute noise.

Of the 6,954 land-dispute cases, only 2,503 have at least one outbound case citation of their own, which is required for graph construction and for verification. We sampled 500 cases from this pool, stratified evenly across decades (roughly 62–65 per decade, 1950s–2020s), preferring more heavily-cited cases within each decade. All 500 sampled judgment PDFs have been downloaded and verified to contain clean, fully digitized text, with no OCR step required.

Inspecting the judgment PDFs surfaced a source of citation polarity we had not planned to rely on: pre-2000 Indian Supreme Court judgments include a HEADNOTE section that states it explicitly, in a recurring pattern — a list of “Case v. Case (Citation)” entries followed by a verb such as relied on, followed, approved (positive), distinguished, disapproved, overruled (negative), or referred to

(neutral). This convention is a hard boundary rather than a declining trend: 0 of 190 sampled 2000–2025 judgments contain a HEADNOTE section at all. Post-2000 sign extraction therefore goes through an LLM classifier that locates the sentence(s) around a citation mention in the judgment body and grounds the surrounding free text into the same treatment classes, since the same distinguishing or overruling is expressed there with no fixed vocabulary (e.g. “We are of the view that the said judgment would not even remotely be applicable to the facts of the present case,” which uses none of the HEADNOTE verbs). Both sources write to the same signed-edge format, so downstream graph construction does not need to know which era a label came from. Merging both sources over the full 500-case sample produced 8,467 candidate citation edges, of which 62.6% carry a real sign label after the merge (33.6% positive, 26.9% neutral, 2.1% negative, with the remainder left unsigned where neither extractor could establish a label).

A. Joining Identifiers to Documents

Nothing in the dataset links Indian Kanoon document identifiers to PDF files across the whole corpus; the land-dispute spreadsheet covers only its own rows. We join them on normalized case titles, which locates 5,462 of the 6,151 cases the citation metadata describes (88.8%). The normalization is not trivial to get right: filenames carry a copy marker after the year (`_on_23_May_1957_1`) while titles do not, so stripping “trailing digits” from both silently removes the year from titles, and every pair then mismatches by exactly that year. This failure is invisible — it produces a join, just a nearly empty one (0.6% in our first attempt). The join therefore asserts its own accuracy against 500 independently verified identifier/filename pairs, achieving 94%, and refuses to write a mapping if that rate falls, because a silent join failure would corrupt every downstream measurement. Judgment text for all joined cases is cached once so that fact segmentation, scrubbing and index construction do not re-parse PDFs.

B. A Measured Ceiling on Retrievable Precedent

Because the archive holds Supreme Court judgments while those judgments cite High Courts, the Privy Council and English decisions, many authorities cannot be retrieved by any method. We measured this rather than assuming it. Of 46,904 case-citation edges, 11,991 (25.6%) point at a case that is present and joinable; of 22,119 distinct cited authorities, only 3,074 (13.9%) are. All recall figures in Section VI are reported against these reachable authorities, with raw totals given, so the ceiling stays visible. 1,206 cases have both their own text and at least three reachable authorities; 771 of those are post-2000 and form the query pool.

IV. Methodology

The pipeline runs in five deterministic stages with a single constrained generative step. A language model

appears only at stage four, and only inside a candidate set it is not permitted to leave.

A. Problem Formulation

Let J be a judgment with factual narrative $F(J)$ and outbound case-citation set $C(J)$ taken from the provided metadata. The task is to predict $C(J)$ from $F(J)$ alone. Because $C(J)$ is recorded independently of the judgment text, labels are free and exist for every case the metadata covers; no manual annotation is required to evaluate retrieval. The haystack is the set of 5,462 joinable judgments, and a candidate may enter the ranking only if its decision year does not exceed that of J — a constraint that is legally necessary and also removes a large class of trivially wrong candidates.

B. Citation Scrubbing

The formulation is only honest if $F(J)$ cannot contain $C(J)$. A judgment names the authorities it relies on in the same sentences that reason about them, so any sentence carrying a case-citation signal — a v./vs. connector, or a law-reporter reference such as i.l.r., s.c.r., air or scc followed by a volume — is removed before the text is used as a query. The filter is deliberately blunt and discards genuine factual content along with the citations, which is far cheaper than leaking a label. We quantify the residual leakage in Section VI-F.

C. Retrieval

Two retrievers sit behind one interface, so both are scored by identical code and the comparison between them is meaningful rather than incidental.

Lexical. A BM25 index ($k_1=1.5$, $b=0.75$) over the full text of every judgment. Document frequencies are computed in a streaming pass and postings stored in typed arrays, since the corpus averages roughly 60,000 characters per judgment. Query terms are capped at the 500 highest $\text{tf} \cdot \text{idf}$ terms, because a whole judgment is a ten-thousand-token query and scoring every term means walking a posting list per term for negligible gain.

Dense. Each judgment is represented by ten chunks sampled evenly across its whole length, embedded with a 4,096-dimensional instruction-tuned embedding model and mean-pooled into one L_2 -normalized vector, so cosine similarity is a single matrix-vector product. Sampling across the document rather than truncating its opening matters: a single embedding of the first page would lose to BM25 for reasons of representation rather than method. The instruction prefix the embedding model expects is applied to queries only, never to documents.

D. Signed Graph and Ranking

Citation edges are signed by merging two complementary sources into one format: a regular-expression extractor over the pre-2000 headnote convention, which yields high-precision labels at no inference cost, and a classifier for modern judgments, which grounds the text

surrounding a citation into the same treatment vocabulary. The result is 8,467 edges of which 62.6% carry a polarity label. A structural proof-support score over this graph, following the modeling intuition of signed graph learning [2], indicates whether surrounding case law presently supports or has weakened a proposition — explicitly not a prediction of how a future court would rule. We flag a limitation immediately: only 174 edges are negative (5 overruled, 17 overruled or disapproved, 141 distinguished), so the “still good law” signal is thin, and proof support derives partly from in-degree, which risks collapsing toward the popularity control of Section V.

E. Certification

Retrieval cannot fabricate: every candidate is a real case drawn from the corpus, so checking membership would be a tautology. The fabrication risk appears at generation, when a model can cite something absent from its brief or garble a name into a case that does not exist. The model receives the ranked candidates and is asked for an IRAC analysis citing only from that list. Each named authority is then resolved back against the candidate set in two stages. First a verbatim match on normalized text, since the model is instructed to copy captions exactly and a faithful copy — including a truncated one — should certify with no fuzziness. Failing that, a party-name matcher tuned against a hand audit, requiring that every usable party of the cited name match some party of a candidate, with a coverage floor so that a single shared generic token cannot carry a match.

Keeping the stages distinct is what makes the numbers interpretable. The matcher is deliberately strict and refuses, for instance, “State of Bombay v. Advani” for “Province of Bombay vs Kusaldas S Advani”. That is correct behavior, but a refusal for sloppy paraphrase would otherwise be indistinguishable from a refusal for outright fabrication. Refusals are therefore further split by whether the cited name resolves to any case in the corpus: one that names a real but unretrieved case is a retrieval miss, whereas one that resolves to nothing is the fabrication signal.

F. Ontology-Directed Extraction, Repositioned

An earlier version of this work made ontology-directed extraction the mechanism that found authorities. A starter ontology defines entity classes (Party, Court, LegalInstrument, PriorCase, PropertyAsset), action classes (Procedural, Dispute, Treatment, Ruling, Disposition), semantic roles (Agent, Patient, Authority, Forum, Time) and target concepts (Claim, Issue, RuleCited, AuthorityCited, AuthorityTreatment, Fact, Holding, Outcome). Symbolic rules are induced from 24 hand-annotated examples in the manner of AutoSlog [6], with LLM grounding replacing Riloff’s syntactic templates: the model maps a sentence onto ontology classes and roles rather than a parse tree,

TABLE I
Recall against reachable authorities, 100 post-2000 query cases

k	Facts only (2,000 ch.)		Whole judgment		
	BM25	Dense	BM25	Dense	Popularity
10	15.6%	16.2%	31.0%	25.9%	5.3%
50	30.4%	33.7%	51.5%	48.5%	14.8%
100	42.1%	41.3%	62.4%	58.7%	23.7%

and a rule fires when a grounded (class, role) pair matches a gold filler.

This machinery is retained — it locates citation spans for scrubbing, produces labelled data, and provides an auditable typed layer — but it is no longer what finds authorities, because measurement showed retrieval does that better. Section VI-G reports its two diagnosed failures and the one that remains open.

V. Experimental Setup

Retrieval is evaluated on 100 post-2000 query cases drawn from the pool of Section III-B, holding 532 reachable authorities out of 1,659 cited (32.1% reachability). Generation is evaluated on 25 cases with $k=20$ candidates each. The generator is a 17B instruction-tuned model and the embedding model has 4,096 dimensions; both run on an institutional inference endpoint, so call volume is not a design constraint.

Query cases are restricted to post-2000 judgments. This is not a convenience: a pre-2000 headnote lists the cited authorities at the top of the document, so a query built from such a judgment contains its own answer even after sentence-level scrubbing. The restriction inverts the difficulty profile of the extraction work, where pre-2000 judgments were by far the easiest.

Two controls are essential. A popularity control always returns the most-cited cases in the corpus, subject to the same date constraint. Citation graphs are strongly skewed, so this is a deceptively strong baseline, and any retriever that cannot beat it is not reading the dispute. A closed-book configuration receives the same facts with no retrieved candidates, isolating what the retrieval-and-certification stages contribute.

VI. Results

A. Precedent Retrieval from Facts

Table I reports recall against reachable authorities. Retrieval from a 2,000-character fact statement recovers 15.6% of reachable authorities at $k=10$, roughly three times the popularity control, rising to 42.1% at $k=100$. Precision at $k=10$ is 8.3% for the fact-only arm and 16.5% for the whole-judgment arm, which is what determines whether the output functions as a usable shortlist rather than a pile.

TABLE II
Authorities proposed in generated answers, 25 cases, $k=20$

Config.	Prop.	Cert.	Refused		Grnd.
			real	none	
Closed-book	61	3	12	46	4.9%
Retrieve, certify	61	61	0	0	100.0%

B. The Facts Carry Most of the Signal

Reducing the query from a whole scrubbed judgment (roughly 60,000 characters) to 2,000 characters of facts — a thirtyfold reduction — costs only half the recall at $k=10$. The court’s reasoning therefore contributes less than expected, which is encouraging for the deployed setting where only facts are available. Intermediate arms at 10,000 and 5,000 characters are within noise of each other at $k=50$ (37.4% and 38.2%), indicating a plateau: there is little value in longer fact statements beyond roughly 5,000 characters.

C. Lexical and Dense Retrieval Trade Places

Dense retrieval leads on short fact queries (33.7% against 30.4% at $k=50$) while BM25 leads on whole judgments (51.5% against 48.5%). The pattern is consistent with what each method exploits: with little text, semantic similarity helps; with a great deal of text, exact overlap on statute names, doctrine names and party names beats a single pooled vector. The gap at $k=10$ is within noise on 532 authorities; the gap at $k=50$ appears real. Since the deployed query is short, dense is marginally the better default, and hybrid fusion [5] is the obvious untried lever.

D. Certification Eliminates Fabrication

Table II is the central result. Given the facts and no retrieved brief, 46 of 61 proposed authorities resolve to no case anywhere in 5,462 judgments — a 75% fabrication rate, in fluent and correctly formatted citations, reproducing on this corpus the failure mode reported by Dahl et al. [1]. Drafting from a retrieved brief and certifying every citation drives this to zero. Because certification is ordinary code rather than a second model judging the first, the guarantee does not depend on a model behaving well.

E. Groundedness Is Not Correctness

Only 13.1% of the 61 certified authorities were authorities the court actually cited. Certification guarantees that an authority is real, present in the corpus and pre-dating the dispute; it does not guarantee that it is the right authority. This is consistent with the precision figures reported above and it is the gap that ranking is aimed at. We regard reporting these two quantities separately as a methodological requirement rather than a caveat: a system that conflated them could claim perfect performance while citing irrelevant cases.

F. Residual Leakage

Sentence-level scrubbing does not remove a judgment’s caption, so party names remain in the query, and Indian litigation frequently runs through repeated appeals between the same parties. We measured the effect: correct hits share a party name with the query at 16.9% against a base rate of 8.5%, worth about 1.7 points of recall at $k=10$ (15.6% falling to 13.9% if every same-party candidate is discarded). That figure is a conservative floor, because the discard also removes legitimate same-matter citations — a later appeal citing the earlier decision in the same litigation is a correct retrieval, not leakage — and because the detector over-flags common surnames and generic descriptors.

G. Two Diagnosed Extraction Failures

The typed-extraction layer was bounded by candidate ordering rather than model capability. Judgments average roughly 515 keyword-matched sentences and only the first 30 were grounded, in document order. Measured decomposition showed that 88.9% of real citation edges have a party name somewhere in the text and 51.1% survive the keyword filter, but only 6.9% fell inside that window — which explains why extraction worked on 1950s judgments, whose headnotes announce citations at the top, and produced almost nothing after 1990. Spending the identical budget on citation-shaped sentences first raised the ceiling to 23.5%. The lesson is methodological: the decomposition required no inference calls and predicted the outcome before any expensive run.

Second, extraction rules keyed only on an ontology class and role, resolved by majority vote. An overloaded class absorbed three target concepts and silently discarded two of them. Keying rules additionally on the content shape of the extracted span — whether it reads as a question, a case name, or a statutory provision in subject position — removed all 196 misrouted statute references from the authority channel with byte-identical verified counts, and populated the Rule element of the IRAC output for the first time.

One gap remains open. The Issue and Fact concepts are still not extracted, because grounding truncates the relevant spans before any rule can apply. We tested whether a larger model resolves this across five candidates and it does not: two larger models scored below the smaller baseline, and although two produced Fact rules, those rules fire on nothing in real text. Rules are also coupled to the grounding model that induced them, since they key on the (class, role) pairs a particular model emits, so substituting the model silently invalidates the rule set. The remaining work is contrastive annotation, not more compute.

VII. Discussion and Limitations

The corpus bounds what any method can achieve here. With 25.6% of citation edges pointing outside the archive, a perfect retriever would still miss three authorities in

four, and adding High Court judgments would raise the ceiling more than any change of method. We regard stating this alongside every recall figure as necessary; a paper reporting the same numbers without it would look considerably better and mean considerably less.

The query representation is a proxy. A fixed-length prefix of scrubbed text is not a fact statement, and no reliable section structure exists to extract one: sampled across six decades, an explicit facts header appeared in none of twelve judgments and issue in two, so the dataset’s claim of structured sections does not hold of the PDFs. A fact segmenter is required before the deployed setting can be claimed faithfully.

Polarity is thin. With 174 negative edges out of 8,467, the signed graph cannot yet support strong claims about whether a precedent remains good law, and proof support overlaps with in-degree, so ranking must be validated as an ablation against the popularity control rather than assumed to help.

Finally, the evaluation is small. One hundred query cases and twenty-five generation cases support the direction of these effects but not fine-grained comparisons, and the 94% identifier join means a minority of labels are wrong in both directions.

VIII. Future Work

Four items follow directly. Wiring proof-support and polarity into ranking is the most direct route to closing the gap between groundedness and correctness, and is cheap because the graph already exists. Hybrid lexical-dense fusion should beat either retriever alone given that they lead under complementary conditions. Contrastive annotation for the Issue and Fact concepts is the remaining extraction work, now that the model-capacity hypothesis has been tested and rejected. Finally, the ontology lacks any notion of an element — the constituent requirements a claim decomposes into, each needing factual support — which is what would carry the pipeline from “these authorities apply” to “these facts satisfy these elements”; the content-guard rule keying is a natural template, since deciding which element a fact satisfies is the same shape of problem.

IX. Conclusion

We inverted a legal extraction pipeline into a precedent prediction pipeline and found that the inversion is both more useful and better supported by evidence. Predicting authorities from facts alone beats a popularity control by roughly threefold, and most of the retrieval signal turns out to live in the facts rather than in the court’s reasoning. The result we regard as the contribution is not a retrieval score but a guarantee: requiring every cited authority to resolve deterministically back to a retrieved case reduces fabricated citations from 75% to zero, and does so without depending on the generating model’s cooperation. We also report what certification does not buy — only 13.1%

of certified authorities were the ones the court chose — because a proof of provenance is not a proof of relevance, and conflating them would misrepresent what a proof-carrying system can promise.

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